

Assignment 4: Chapters 18-22

Due 5pm, 6 May 2026

Answer each of the following questions. Amongst the three questions, the total amount of writing should not exceed 500 words.

1. Briefly describe the philosophical arguments in favor of Ptolemaic (geocentric) cosmology and against Copernicus' model from the perspective of one of Copernicus' earliest readers.
2. Briefly describe the Aristotelian argument against the earth's movement. Describe Tycho Brahe's argument against the moving earth. Explain which of these two is refuted by Galileo's moving boat experiment.
3. This course teaches us that a model of nature based on incorrect concepts can lead to empirically adequate predictions of nature. Adequate empirical predictions are the basis of belief in current models/theories of nature. How can we be sure that the *concepts* that current models are based on are correct?